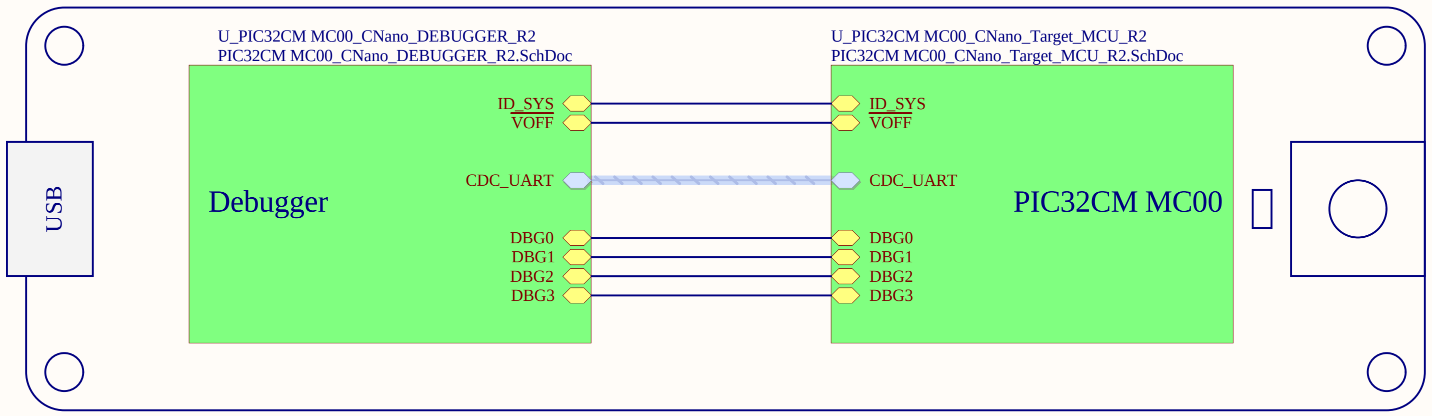
PIC32CM MC00 Curiosity Nano



Program / Debug Interface	
SWD	SWD

IDE

Program   Debug

Mass Storage

CURIOSITY (E:)

Serial / CDC port	
USART3 TX	PA01
USART3 RX	PA00

Terminal

> Hello World!

Data Visualizer

DGI (Data Gateway Interface)	
Debug GPIO 0	PA31
Debug GPIO 1	PA30

U_PIC32CM MC00_CNano_Rev_History_R2
PIC32CM MC00_CNano_Rev_History_R2.SchDoc

A09-3392: Product
- A08-2804: Kit Box
- A08-2996: Outer Label
- *A09-3304: ESD Bag w/Hdrs
  - A08-1664 ESD Bag
  - A08-3031 (2) 1x17pin Hdrs
  - A09-0614: Inner Assembly Label
  - *A09-3413: ESD Bag w/Pop Board and label
    - A09-0608: ESD bag
    - A09-0614: Inner Assembly Layer
    - **PCBA: A09-3389 (Populated PCB w/label)**
      - **A08-3091: bare PCB**
      - Axx-xxxx: all electronic components
      - A09-0611: Label PCBA
      - A11-0406: nEDBG firmware
      - A11-0496 PIC32CM MC00 CNANO Test
      - A12-1314: CNANO Test Instructions
      - A12-1463: PCBA docs

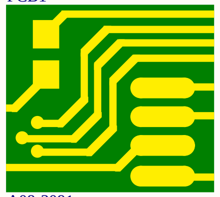
TESTDOC1
A12-1314

TEST1
$ >_
A11-0496

FW1
A11-0406

PCBADOC1
A12-1463

PCB1



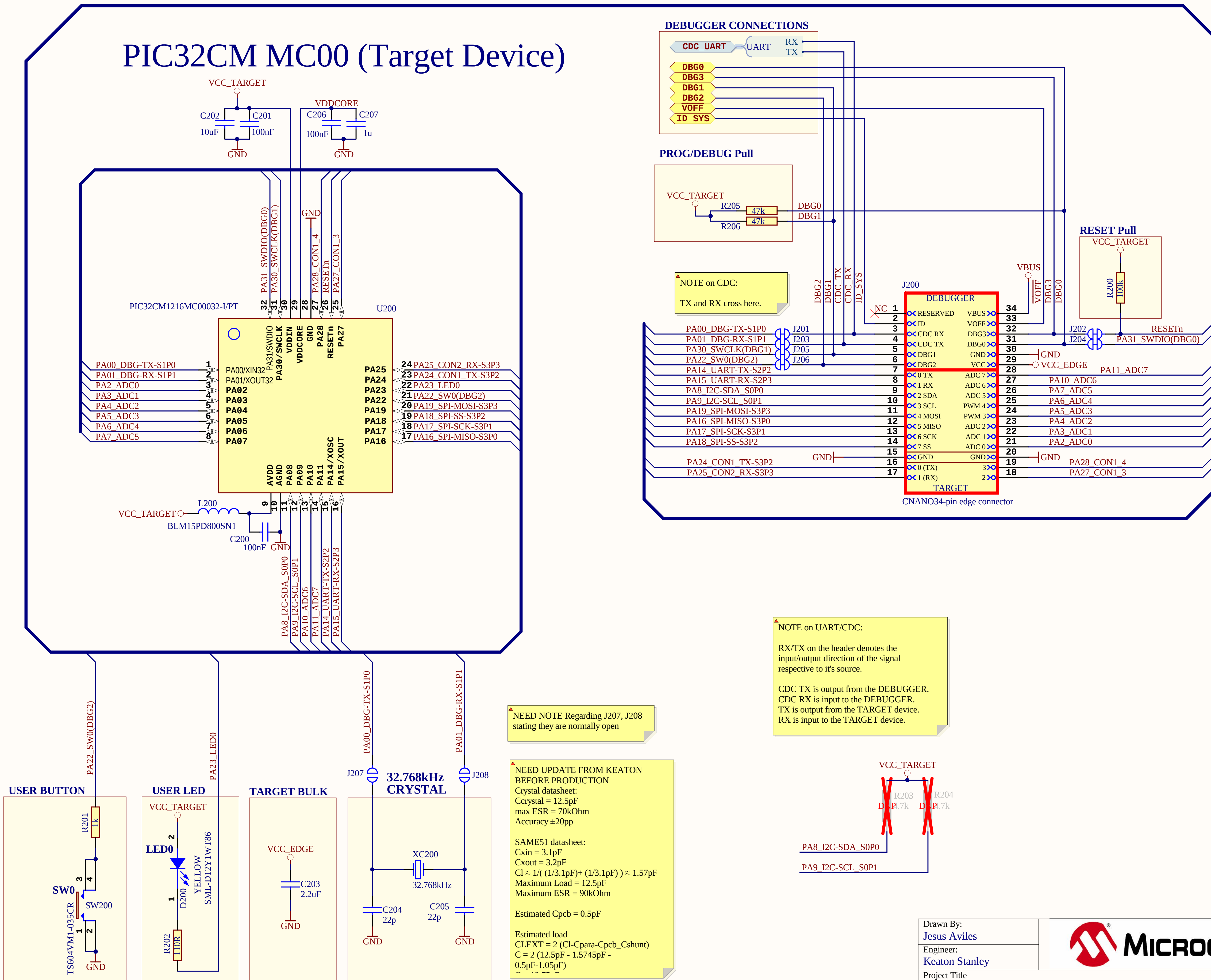
A08-3091

LABEL1: A09-0611

Product number/revision
Serial number

Label PCBA

Drawn By: Jesus Aviles		MICROCHIP	
Engineer: Keaton Stanley			
Project Title: *PIC32CM MC00 Curiosity Nano*			Designed with Altium Altium.com
Sheet Title: Top Level			
Size A3	PCB Assembly Number: A09-3389	PCBA Revision: 2	
	PCB Number: A08-3091	PCB Revision: 2	Date: 9/17/2020
File: PIC32CM MC00_CNano_TopLevel_R2.SchDoc			Page: 1 of 4



Drawn By: Jesus Aviles

Engineer: Keaton Stanley

Microchip

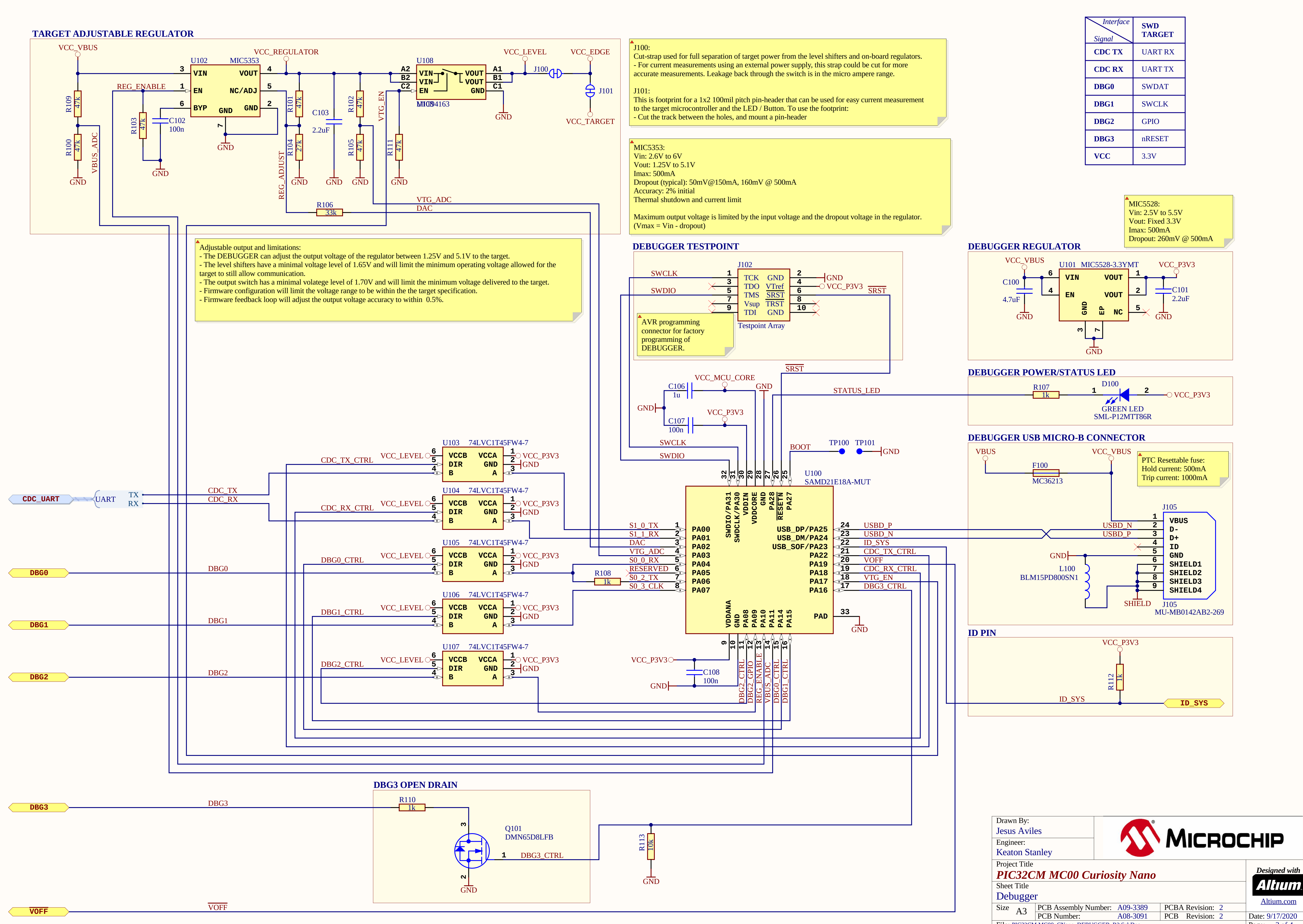
Project Title: *PIC32CM MC00 Curiosity Nano*

Sheet Title: Target MCU

Designed with Altium — Altium.com

Size	PCB Assembly Number: A09-3389	PCBA Revision: 2	Date: 9/17/2020
A3	PCB Number: A08-3091	PCB Revision: 2	Page: 2 of 4

File: PIC32CM MC00_CNano_Target_MCU_R2.SchDoc



J100:
Cut-strap used for full separation of target power from the level shifters and on-board regulators.
- For current measurements using an external power supply, this strap could be cut for more accurate measurements. Leakage back through the switch is in the micro ampere range.

J101:
This is footprint for a 1x2 100mil pitch pin-header that can be used for easy current measurement to the target microcontroller and the LED / Button. To use the footprint:
- Cut the track between the holes, and mount a pin-header

MIC5353:
Vin: 2.6V to 6V
Vout: 1.25V to 5.1V
Imax: 500mA
Dropout (typical): 50mV@150mA, 160mV @ 500mA
Accuracy: 2% initial
Thermal shutdown and current limit

Maximum output voltage is limited by the input voltage and the dropout voltage in the regulator. (Vmax = Vin - dropout)

Adjustable output and limitations:
- The DEBUGGER can adjust the output voltage of the regulator between 1.25V and 5.1V to the target.
- The level shifters have a minimal voltage level of 1.65V and will limit the minimum operating voltage allowed for the target to still allow communication.
- The output switch has a minimal volatege level of 1.70V and will limit the minimum voltage delivered to the target.
- Firmware configuration will limit the voltage range to be within the the target specification.
- Firmware feedback loop will adjust the output voltage accuracy to within 0.5%.

AVR programming connector for factory programming of DEBUGGER.

Signal \ Interface	SWD TARGET
CDC TX	UART RX
CDC RX	UART TX
DBG0	SWDAT
DBG1	SWCLK
DBG2	GPIO
DBG3	nRESET
VCC	3.3V

MIC5528:
Vin: 2.5V to 5.5V
Vout: Fixed 3.3V
Imax: 500mA
Dropout: 260mV @ 500mA

PTC Resettable fuse:
Hold current: 500mA
Trip current: 1000mA

Drawn By: Jesus Aviles
Engineer: Keaton Stanley
Project Title: PIC32CM MC00 Curiosity Nano
Sheet Title: Debugger
Size: A3
PCB Assembly Number: A09-3389 | PCBA Revision: 2
PCB Number: A08-3091 | PCB Revision: 2
File: PIC32CM MC00_CNano_DEBUGGER_R2.SchDoc
Designed with Altium.com
Date: 9/17/2020
Page: 3 of 4

Revision History

PCB Assembly Rev 1:

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

Design Changes:

PA31_SWDIO(DBG0) & PA22_SW0 (DBG2) - swapped

PCB:

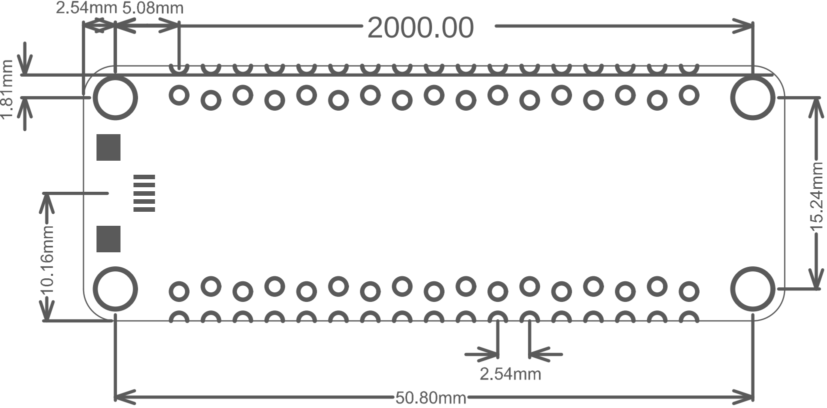
PCB Assembly Rev 3:

Design Changes:

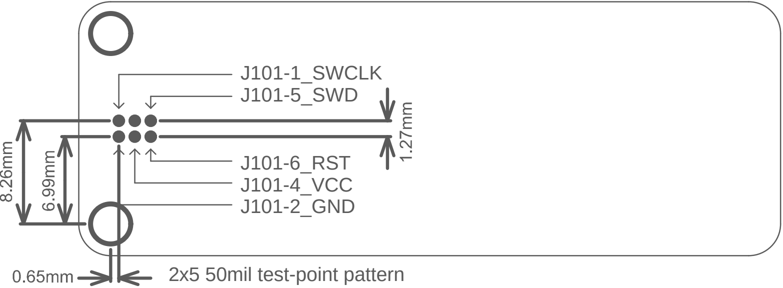
PCB:

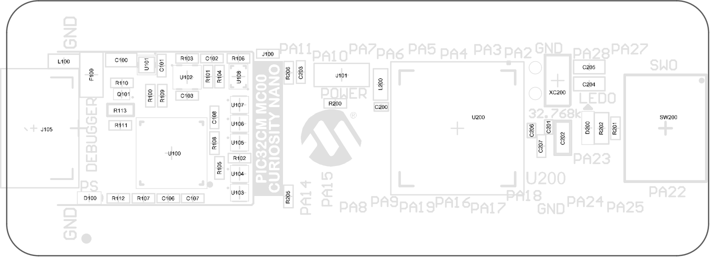
Drawn By: Jesus Aviles		
Engineer: Keaton Stanley		
Project Title: PIC32CM MC00 Curiosity Nano		Designed with Altium
Sheet Title: Revision History		Altium.com
Size: A3	PCB Assembly Number: A09-3389	PCBA Revision: 2
	PCB Number: A08-3091	PCB Revision: 2
		Date: 9/17/2020
File: PIC32CM MC00_CNano_Rev_History_R2.SchDoc		Page: 4 of 4

Connector Placement



Test Point Placement





PIC32CM MC00
CURIOSITY NANO
DEBUGGER
PS
GND
PA11
PA10
PA7
PA6
PA5
PA4
PA3
PA2
PA28
PA27
SWO
POWER
LED0
32.768k
PA23
U200
PA22
PA14
PA15
PA8
PA9
PA19
PA16
PA17
PA18
PA24
PA25
J105
L100
F100
C100
U101
C101
R103
C102
R106
J100
R206
C203
J101
L200
XC200
C205
C204
SW200
R110
Q101
U102
R101
R104
U108
R100
R109
C103
R200
C200
U200
R113
R111
U107
C108
U106
U100
R108
U105
R102
R105
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R205
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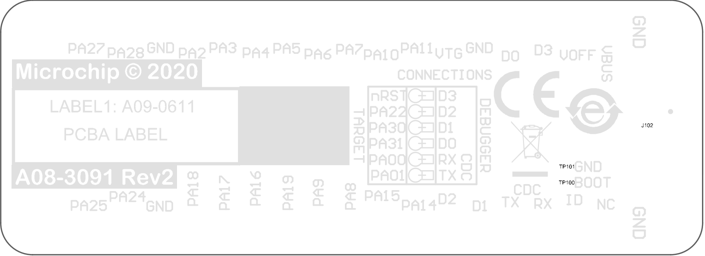
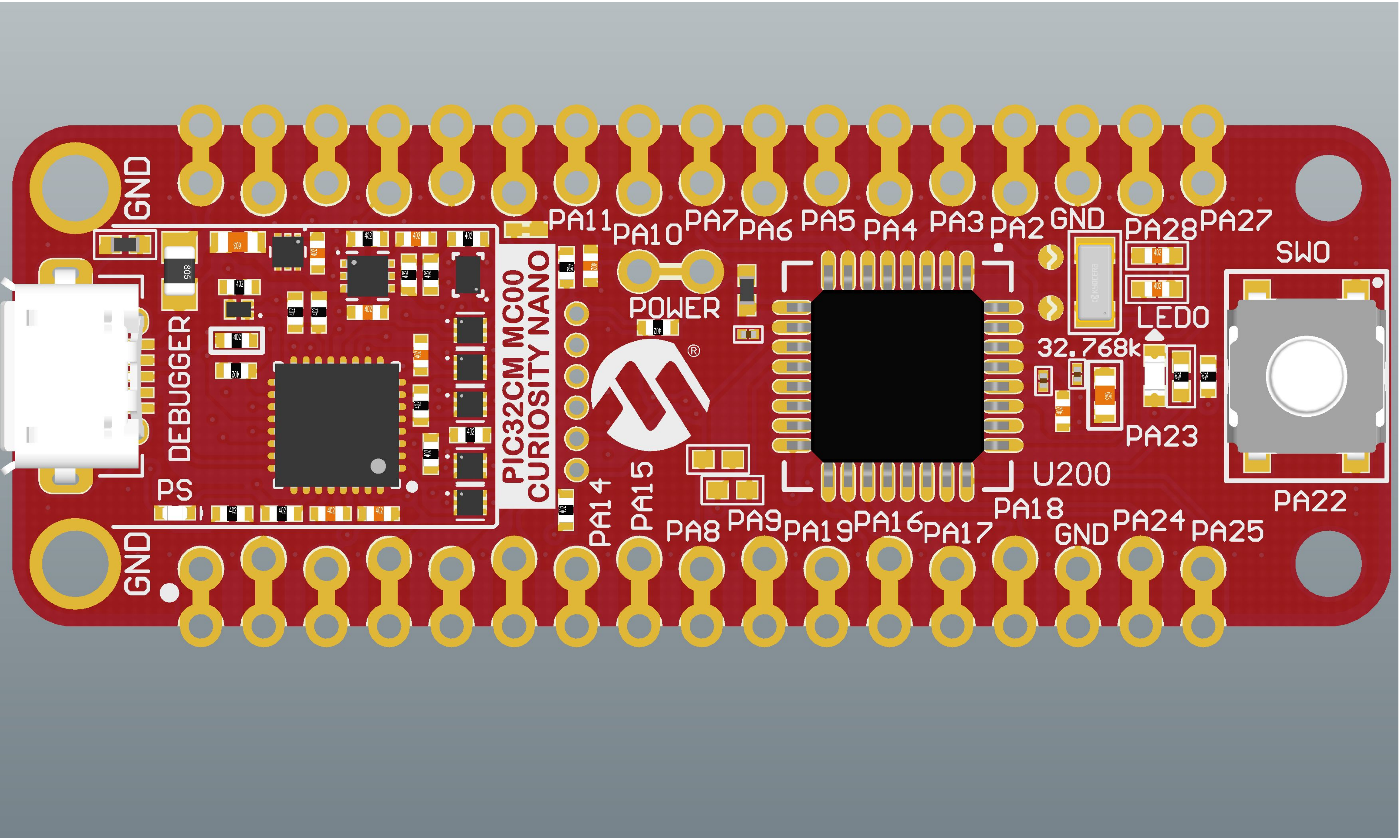
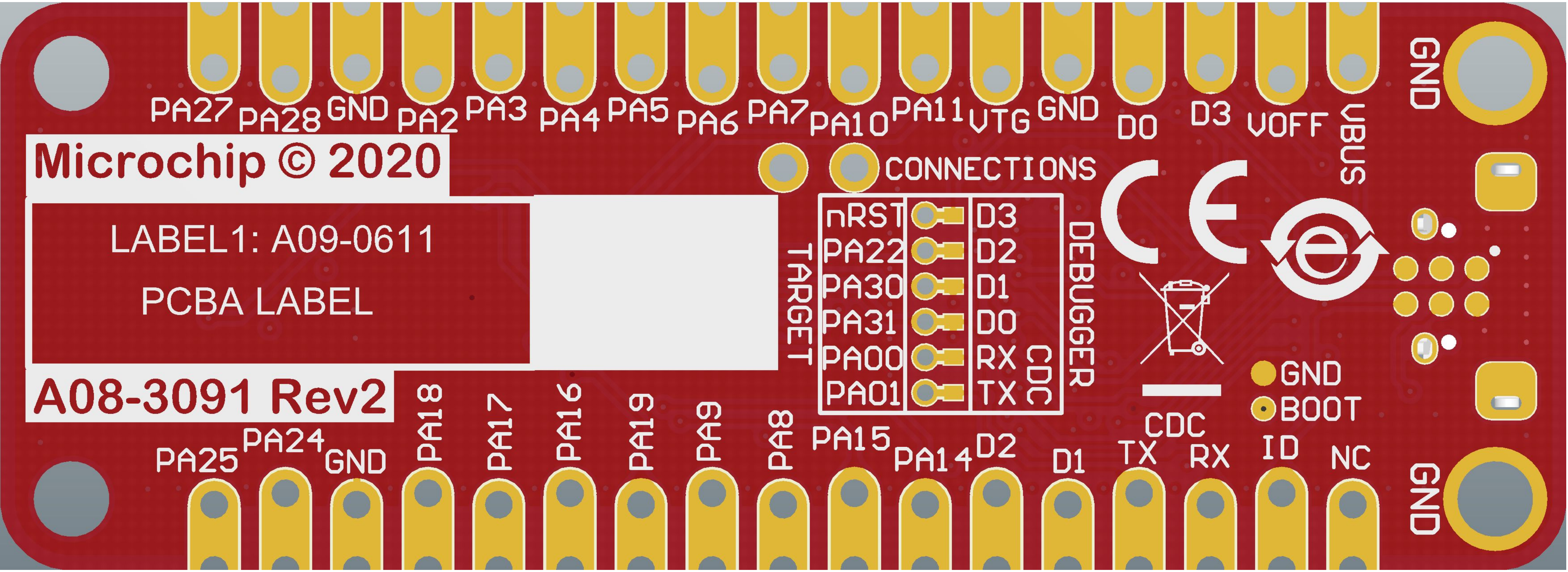
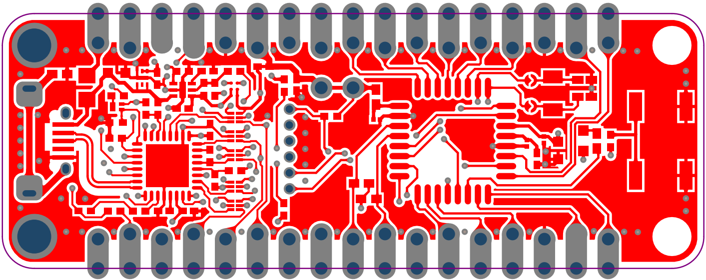
C201
C202
C207
D200
R202
R201


PA27 PA28 GND PA2 PA3 PA4 PA5 PA6 PA7 PA10 PA11 VTG GND
D0 D3 VOFF VBUS GND
Microchip © 2020
LABEL1: A09-0611
PCBA LABEL
A08-3091 Rev2
CONNECTIONS
TARGET
nRST PA22 PA30 PA31 PA00 PA01
D3 D2 D1 D0 RX TX
CDC
DEBUGGER
J102
TP101 GND
TP100 BOOT
PA25 PA24 GND PA18 PA17 PA16 PA19 PA9 PA8 PA15 PA14 D2 D1
CDC
TX RX ID NC
GND

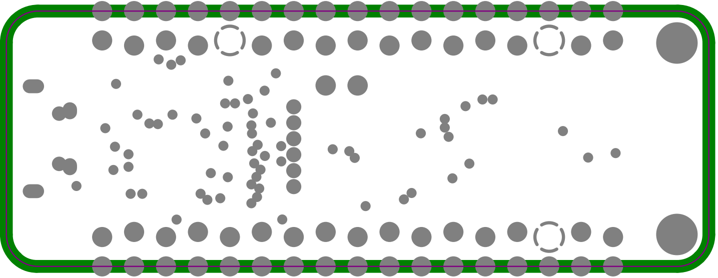

GND
PA11
PA10
PA7
PA6
PA5
PA4
PA3
PA2
GND
PA28
PA27
SWO
POWER
LED0
32.768k
PA23
U200
PA22
DEBUGGER
PIC32CM MC00
CURIOSITY NANO
PS
GND
PA14
PA15
PA8
PA9
PA19
PA16
PA17
PA18
GND
PA24
PA25


PA27 PA28 GND PA2 PA3 PA4 PA5 PA6 PA7 PA10 PA11 VTG GND D0 D3 VOFF VBUS
GND
Microchip © 2020
CONNECTIONS
LABEL1: A09-0611
PCBA LABEL
TARGET
nRST
PA22
PA30
PA31
PA00
PA01
D3
D2
D1
D0
RX
TX
CDC
DEBUGGER
GND
BOOT
A08-3091 Rev2
PA25 PA24 GND PA18 PA17 PA16 PA19 PA9 PA8 PA15 PA14 D2 D1 TX RX ID NC
CDC
GND

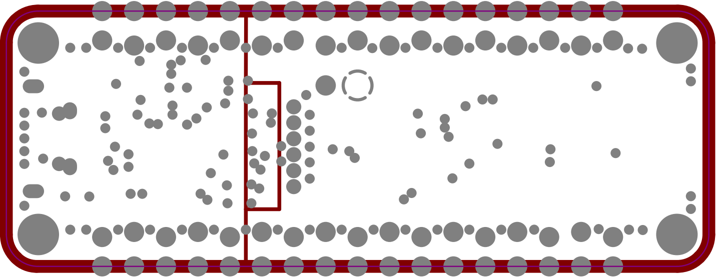
Top Layer



GND Plane



PWR Plane



Bottom Layer